

Ivan Evdokimov, Ph.D.

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Dear Dr Papastylianou,

I am writing to apply for the Senior Research Associate position (Ref: REQ10196) in the School of Computer Science and Electronic Engineering, working on the Innovate UK-funded project with Colchester Borough Homes to identify and support children and households at risk of housing insecurity.

I completed my PhD in Computational Finance at the University of Essex, where I developed a transfer learning framework for financial forecasting, first-authored a peer-reviewed journal publication, and released an open-source package with external adoption. Since 2023, as a Data Engineer at the UK Data Service, I have applied machine learning to a closely related problem: identifying disclosure risk in sensitive social science and socioeconomic datasets.

In my current and previous roles, I have built the kind of end-to-end analytical pipeline this post requires. I trained and deployed classification models (PyTorch, scikit-learn, BERT) for automated metadata classification and disclosure risk assessment in line with ONS Statistical Disclosure Control methodology, achieving 95%+ F1 accuracy on sensitive datasets, and integrating and linking training data from multiple external sources. I used Shapley values to validate feature-level contributions and ensure model interpretability, and applied DBSCAN clustering to segment survey studies by concentration of disclosive household attributes. I also built ETL pipelines to extract, clean, and query PostgreSQL-hosted socioeconomic data at scale, and deployed automated ML retraining pipelines on AWS with drift monitoring, reducing manual intervention.

Beyond the technical work, I have experience communicating findings to non-technical stakeholders and collaborating across teams: as a Senior Research Assistant, I built CI/CD pipelines supporting a multi-team SaaS platform and developed a mobile application used daily by field technicians. I also bring direct experience working within research ethics and data protection frameworks, having handled sensitive personal data throughout my time at the UK Data Service. Through personal projects with a TypeScript React frontend, I have gained hands-on experience in frontend development and shipping production-ready applications — skills directly applicable to building scalable, reliable decision-support systems.

I am particularly drawn to this role because it applies rigorous, interpretable machine learning to a problem with immediate social impact — helping identify vulnerable households before a housing crisis disrupts a child's education and wellbeing. I would welcome the opportunity to discuss how my background in disclosure risk assessment, clustering, and explainable AI could contribute to this project's success.

Kind regards,
Ivan Evdokimov